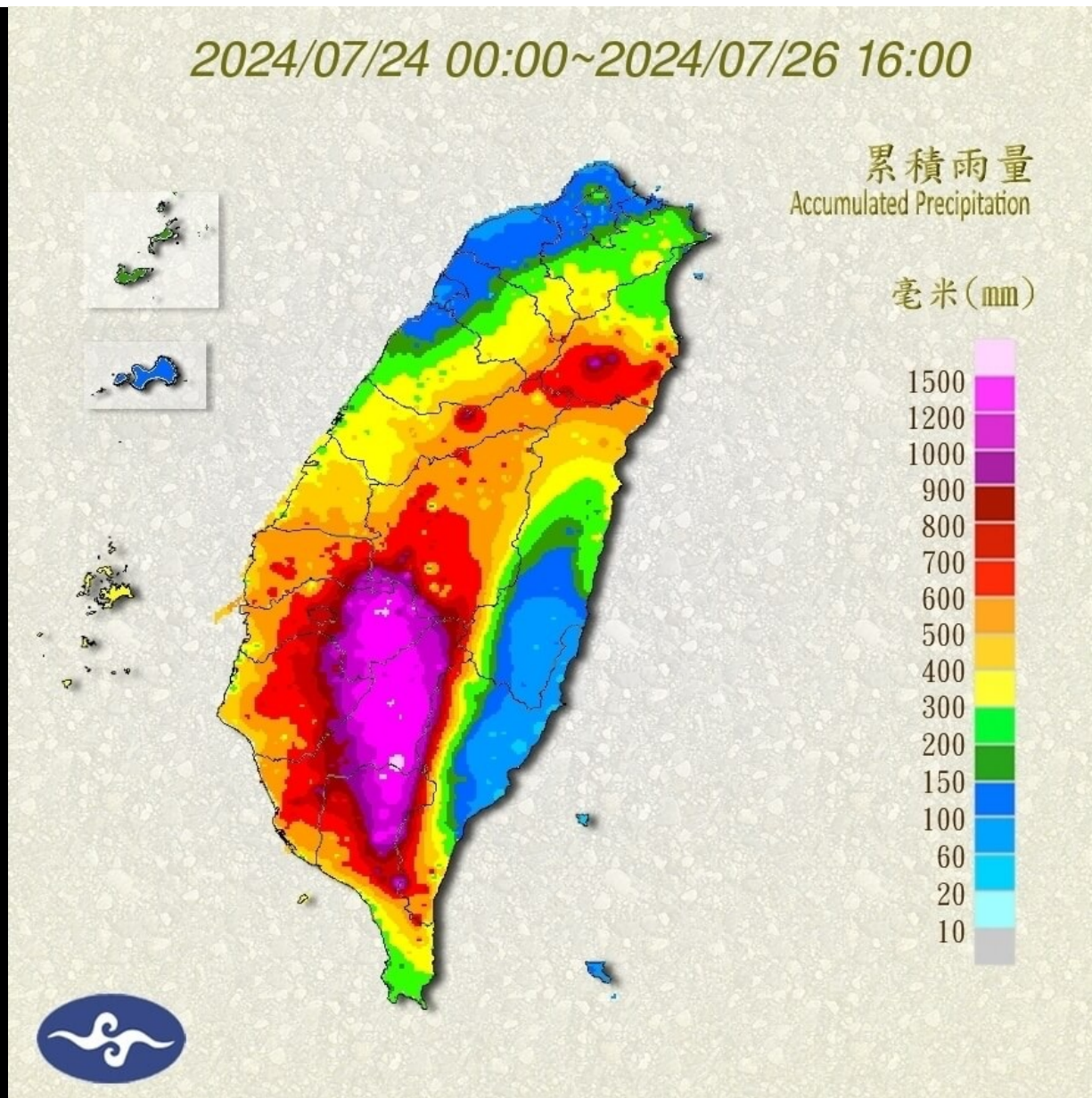


Spatial Interpolation

GEOG 1026
Department of Geography
National Taiwan University



Fundamental Geo-Processing and GIS Analysis

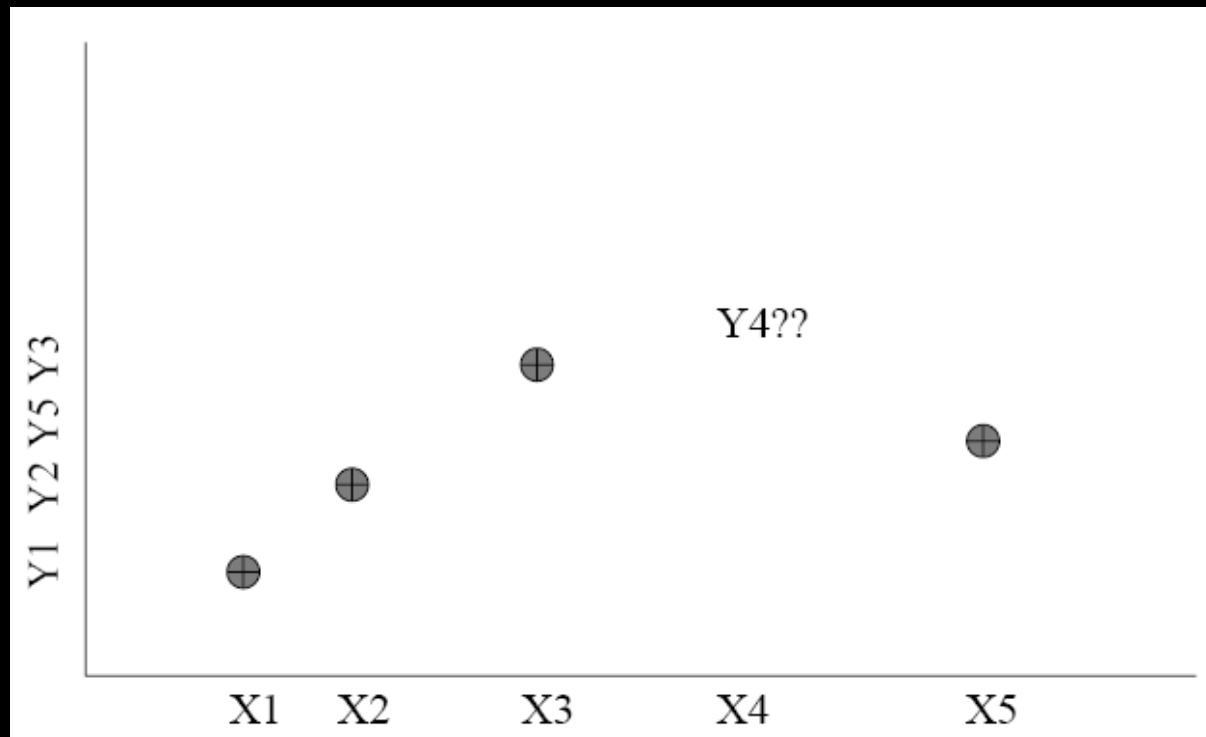
- Spatial and attribute query
- Data transformation
- Data management
- Spatial extraction
- Spatial overlay
- Proximity analysis
- Spatial statistics analysis
- Data classification
- Spatial interpolation
- Terrain analysis
- Network analysis

Fundamental Geo-Processing and GIS Analysis

- Spatial and attribute query
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- **Spatial interpolation**
- Terrain analysis
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What Is Interpolation?

- A method to estimate the values of unknown points by the values of sampled points.



Spatial Interpolation

- A computational procedure of estimating the surface values for a continuous geo-spatial variable (usually interpolated into raster) at unmeasured locations within the area where a set of surface values is given.
- Points close together in space will more likely have similar values than points far apart.

2024/07/24 00:00~2024/07/26 16:00

累積雨量
Accumulated Precipitation

毫米(mm)

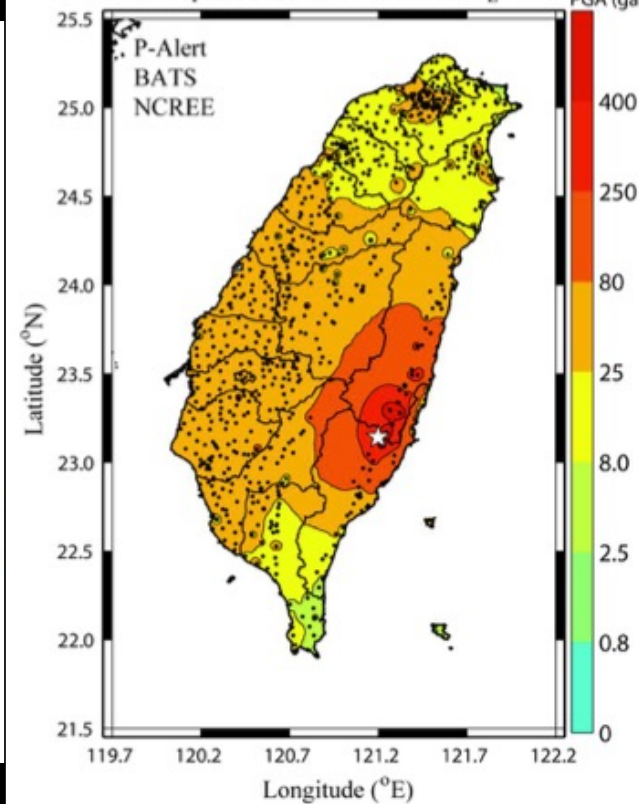
1500
1200
1000
900
800
700
600
500
400
300
200
150
100
60
20
10

Point locations of ozone
monitoring stations

Interpolated prediction
surface

Earthquake 2022/09/18 06:44:15 $M_L=6.8$

PGA (gal)



Interpolation Methods

- Inverse Distance Weighted (IDW)
- Kriging
- Natural neighbor
- Spline / barrier spline
- Topo to raster
- Trend

The Nature of Geographic Data: Spatial Autocorrelation

- Tobler's first law of geography:
 - Everything is related to everything else, but near things are more related than distant.
- Spatial data exhibit an increasing range of values, hence increased heterogeneity with increased distance.
- Spatial autocorrelation measures attempt to deal simultaneously with similarities in the location of spatial objects and their attributes.

IDW (反距離加權法)

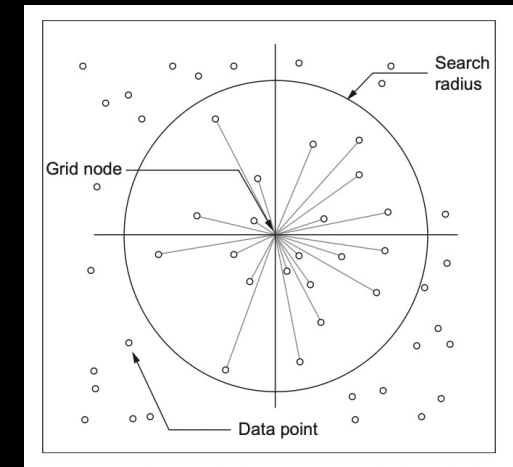
- IDW estimates a raster surface by averaging the values of sample data points in the neighborhood of each processing cell.
- Within searching area, the closer a point is to the center of the cell being estimated, the more influence, or weight, it has in the averaging process.

$$u(x) = \frac{\sum_{i=1}^n w_i(x) u(x_i)}{\sum_{i=1}^n w_i(x)}$$

For normalization (正規化)

$$w_i = \frac{1}{d(x_i, x)^p}$$

- u is data value
- n is total number of known points
- w_i is weighting (權重)
- d is distance
- x is unknown (interpolated) point
- x_i is known (data) point
- p is a positive real number, called power parameter (距離衰減參數)

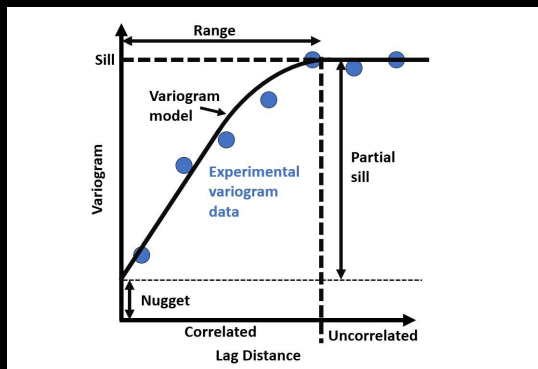


Kriging (克利金法)

- Kriging is an advanced geostatistical procedure that generates an estimated surface from a scattered set of points with z-values.
- More so than other interpolation methods, a thorough investigation of the spatial behavior of the phenomenon represented by the z-values should be done before you select the best estimation method for generating the output surface.

$$u(x) = \sum_{i=1}^n \lambda_i u(x_i)$$

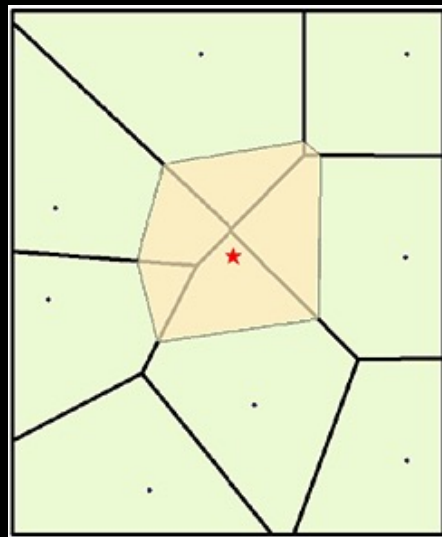
- λ is an unknown weight for the measured value at the i th location.



The weights are based upon the distance between the measured points and the prediction location and also on the overall spatial arrangement of the measured points. The spatial autocorrelation must be quantified to be used in the spatial arrangement of weights. Hence in ordinary kriging the weight, λ , depends on a fitted model to the measured points, the distance to the prediction location, and the spatial relationships among the measured values around the prediction location.

Natural Neighbor (自然鄰域法)

- The method finds the closet subset of input samples to a query point and applies weights to them based on proportionate areas to interpolate a value.
- It is also known as Sibson or “area-stealing” interpolation.
- The proportion of overlap between the new polygon (beige) created around the interpolation point (red star) and the initial polygons (olive) is then used as the weights.



$$u(x) = \sum_{i=1}^n w_i u(x_i) \quad w_i = \frac{A(x_i)}{A(x)}$$

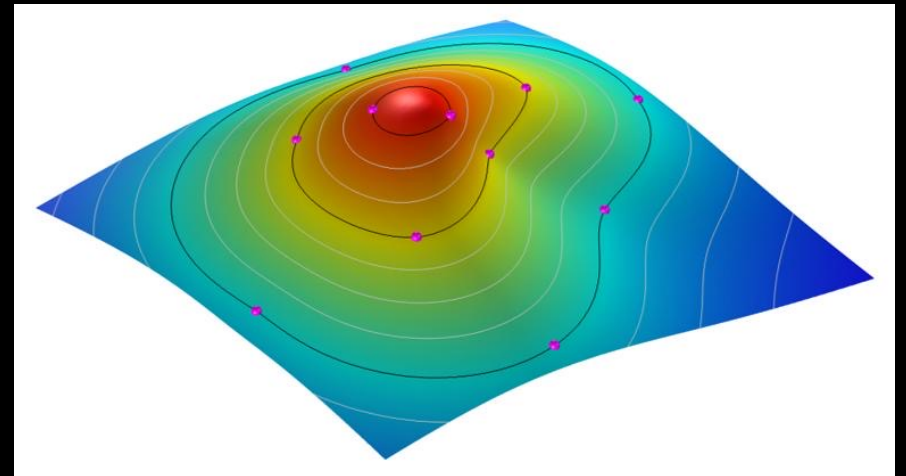
- $A(x_i)$ is the volume of intersection between the new polygon centered in x and the old polygon centered in x_i .
- $A(x)$ is the volume of the new polygon centered in x .

Spline (樣條法)

- Spline estimates values using a mathematical function that minimizes overall surface curvature, resulting in a smooth surface that **passes** exactly through the input points.

$$u(x) = \sum_{i=1}^n \lambda_i \phi(\|x - x_i\|) + p(x)$$

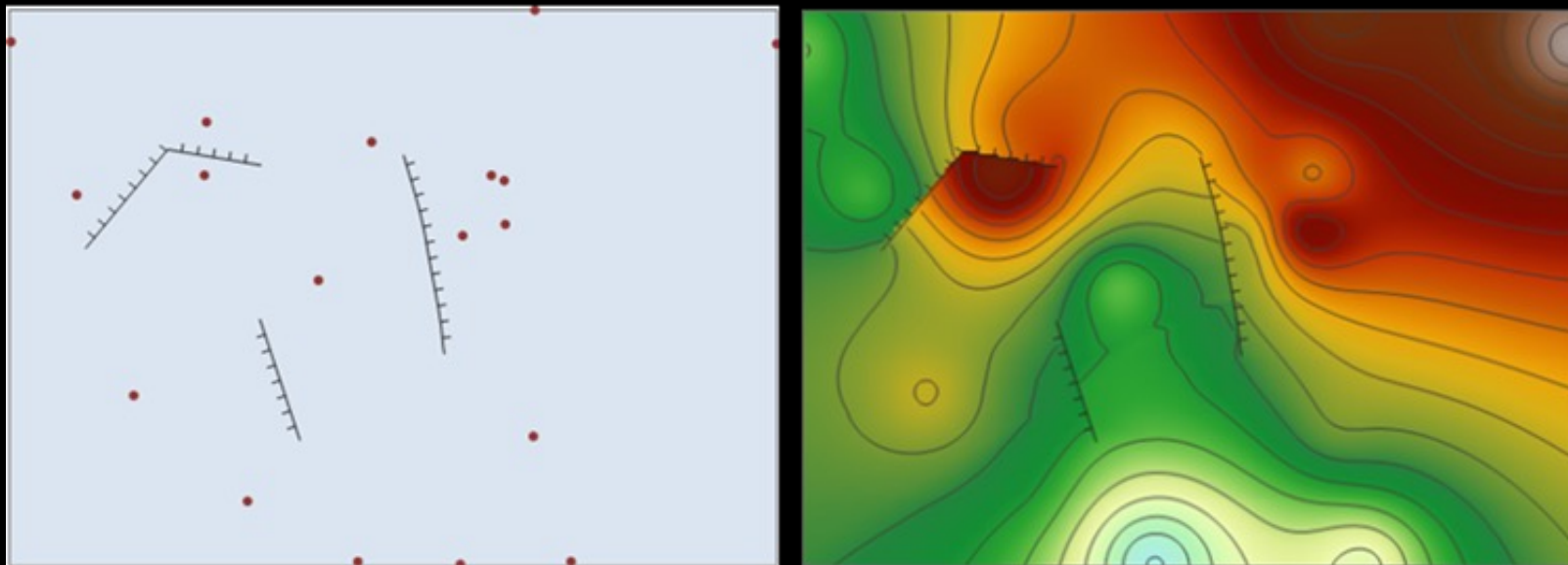
- λ is an unknown weight to be solved
- $\phi(r)$ is a radial basis function, and for thin-plate spline it is usually:
$$\phi(r) = r^2 \ln(r)$$
- $p(x)$ is a polynomial to keep the solution steady



Use data points to form a system of equations (聯立方程式)

Barrier Spline (障礙物樣條法)

- Barrier spline interpolation generates a raster surface using obstacles to interpolate points.



Trend (趨勢面法)

- Trend is a global polynomial interpolation that **fits** a smooth raster surface defined by a mathematical function (a polynomial) to the input sample points.

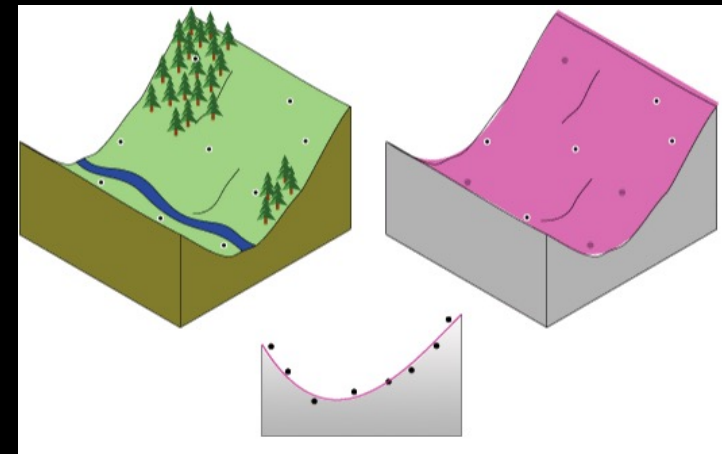
$$u(x, y) = a_0 + a_1x + a_2y + a_3x^2 + a_4xy + a_5y^2 + \dots$$

- $a_0 \dots a_n$ are coefficient

Suppose you have 5 data points $(x_1, y_1) \dots (x_5, y_5)$ with data values $u_1 \dots u_5$

$$\begin{cases} u_1(x, y) = a_0 + a_1x_1 + a_2y_1 + a_3x_1^2 + a_4x_1y_1 + a_5y_1^2 \\ u_2(x, y) = a_0 + a_1x_2 + a_2y_2 + a_3x_2^2 + a_4x_2y_2 + a_5y_2^2 \\ u_3(x, y) = a_0 + a_1x_3 + a_2y_3 + a_3x_3^2 + a_4x_3y_3 + a_5y_3^2 \\ u_4(x, y) = a_0 + a_1x_4 + a_2y_4 + a_3x_4^2 + a_4x_4y_4 + a_5y_4^2 \\ u_5(x, y) = a_0 + a_1x_5 + a_2y_5 + a_3x_5^2 + a_4x_5y_5 + a_5y_5^2 \end{cases}$$

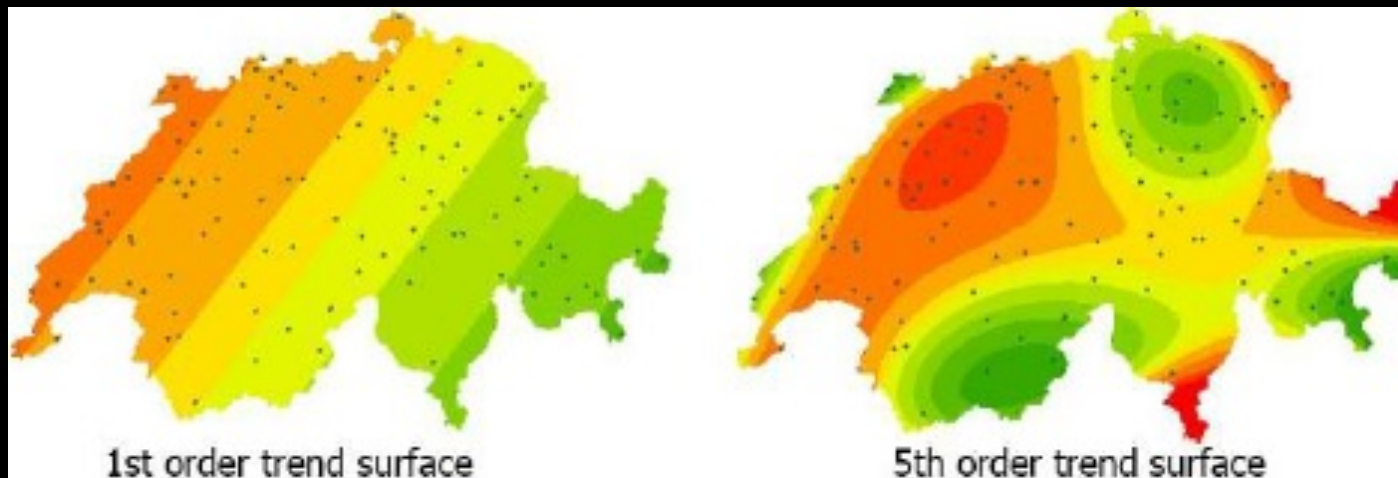
$$\underbrace{\begin{bmatrix} u_1 \\ \vdots \\ u_5 \end{bmatrix}}_d = \underbrace{\begin{bmatrix} 1 & x_1 & y_1 & x_1^2 & x_1y_1 & y_1^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_5 & y_5 & x_5^2 & x_5y_5 & y_5^2 \end{bmatrix}}_G \underbrace{\begin{bmatrix} a_0 \\ \vdots \\ a_5 \end{bmatrix}}_m$$



Least-squared (LSQ) solution:
 $m = (G^T G)^{-1} G^T d$

Trend (趨勢面法)

- The trend surface changes gradually and captures coarse-scale patterns in the data.
- As the order of the polynomial is increased, the surface being fitted becomes progressively more complex. A higher-order polynomial will not always generate the most accurate surface; it is dependent on the data.



Topo to Raster

- The Topo to Raster interpolation is specifically designed to create a surface that more closely represent a natural drainage surface and better preserve both ridgelines and stream networks from different types of input data (elevation points, contour lines, stream lines, and lake polygons).
- The method is a discretized thin-plate spline technique wherein the roughness penalty has been modified in order to allow the fitted DEM (數值地形模型) to capture abrupt changes in terrain.

